

Optimal imperfect maintenance cost analysis of a two-component system with failure interaction

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Abstract

This paper considers the statistical inference in presence of imperfect corrective maintenance for a gradually deteriorating system. According to available data, two inference cases are considered, deterioration based or age based inference. The likelihood function for each of the cases is developed. An application of the inference results for replacement planning is proposed. Numerical implementations give an example of the proposed replacement maintenance planning.

Keywords: Deterioration, gamma process, imperfect maintenance, virtual degradation, virtual age, maximum likelihood method, replacement planning.

Glossary

- X_i The deterioration level just before the i th maintenance, $X_i > L$,
- x_i The realisation of X_i ,
- R_i The improvement factor at the i th maintenance,
- V_i The deterioration level just after the i th maintenance, $V_i < L$, known as virtual deterioration, The realisation of V_i ,
- v_i The realisation of V_i ,
- V_i^* The virtual age after the maintenance

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- v_i^* The realisation of V_i^* ,
- $Y(t)$ The cumulative deterioration at then end of the interval $[0, t]$,
- L The failure threshold,
- T_L The first hitting time of L by a gamma process,
- F_{T_L} The cumulative distribution function of T_L ,
- f_{T_L} The probability distribution function of T_L ,
- T_i The time interval between the $i - 1$ th and i th maintenance
- t_i The realisation of T_i ,

1. Introduction

Nowadays, in presence of high technology sensors and data analyses, complex systems are more often monitored and their health condition analysed. Complex systems are subject to degradation and their health condition deteriorates in time. It is not possible to monitor the whole system but follow some health or degradation indicators. These indicators permit to make decisions on maintenance of the system. In order to make sensible maintenance decisions, it is important to be able to predict the future evolution of the system. The prediction allows to decide whether the system should be maintained correctively or preventively. For such predictions and therefore maintenance decisions, the degradation of the system should be modelled according to expert knowledge or historical data. The development of a mathematical model accurately describing the system's behaviour is essential. However, to be useful, the chosen model should be credible, incorporate the major features of collected data and evolve with time. Health or degradation indicators, are usually influenced by several factors, which are not necessarily known, and their impacts cannot be modelled through a deterministic expression.

In the literature, initially path-wise models are used to describe the degradation phenomenon (see [1, 2]). These models focus more on the average behaviour of the degradation. Later, stochastic processes which are more complex and exhaustive models are proposed, for instance, one can enumerate Gamma processes, Markov chains, counting processes, birth processes

and Wiener processes commonly used for prognostics and failure analysis [3, 4]. Frequently, gamma and Wiener processes are considered to model the evolution of gradual degradation for monotone [5] and non-monotone phenomenon respectively [6]. In the framework of this paper, one adopts for gamma process as degradation model. Once the behaviour of a deteriorating system is modelled, its future failure time can be estimated and it is possible to plan efficient maintenance rules.

Maintenance decision rules and maintenance data analysis are important issues getting more focus on lately. For complex systems where failure can be very costly and have disastrous consequences maintenance policies are of great interest. Maintenance actions can be classified in two major categories: corrective and preventive. Corrective maintenance occurs only when the system fails. In order to avoid this latter, preventive maintenance is carried out when the system is operating. Preventive maintenance can be scheduled according to the age or calendar or be condition-based. In the frame of this paper, we will focus on condition-based maintenance policies and more precisely on policies based on control limit decision rules where the maintenance operation is triggered if the monitored health condition indicator exceeds a critical threshold, refer to [7, 8, 9, 10, 11, 12, 13]. Such policies are different with periodic maintenance. Condition-based maintenance decision rule is based on the information available through on-line monitoring. As control systems and on-line monitoring methods are getting more sophisticated, new on-line information is getting available. It is then important to exploit to the maximum the available on-line information in a relevant process.

In this study, the maintenance of a deteriorating system, such as a large structure, is assumed to be imperfect, i.e. after maintenance, the system is not restored perfectly. For instance, steel structures such as tanks or bridges, are covered by an organic coating in order to protect them from corrosion due to outdoor conditions. The coating layer deteriorates in time, maintenance action are carried out improve the condition of the coating. Mostly maintenance actions are performed in a specific parts and lead to an imperfect maintenance, refer to [14]. Therefore, if the system is not completely replaced, maintenance operations on such systems are mainly imperfect. The main concern in imperfect maintenance policies is to model the impact of maintenance actions on the system. There exist many possibilities to consider imperfect maintenance, refer to [15, 16, 17, 18, 14, 19, 20].

The paper considers imperfect repair based on health indicator and monitoring data. Imperfect repair models frequently use virtual age models [21].

The principle is that the wear-out does not depend on the real age of the system, but on a virtual age, commonly between zero and the elapsed time since the system was new. A virtual age model is completely defined by the failure rate of a new system and by the virtual age hypotheses. Kijima [22] has proposed two widespread classes of virtual age models where the repair efficiency is a random variable supported on the interval $[0, 1]$. Kijima Type I model is such that a repair rejuvenates the virtual age of a proportional amount of the last inter-failure duration, whereas Kijima Type II model supposes that the rejuvenated amount is proportional to the virtual age just before the repair. Some extensions of these models have been proposed in the literature [23, 24, 25, 26, 23, 27, 28, 29, 21, 30, 31, 32, 33].

The imperfect maintenance operation can modify the failure rate of the system, this method is called "improvement factor method" introduced by [18]. The failure rate is modified so that the system is between "as good as new" and "as bad as old" after maintenance. The degree of improvement in failure rate is called improvement factor. This concept is enriched with expert judgment in [34]. Authors in [16] propose a cumulative damage shock model with imperfect periodic maintenance actions reducing the deterioration level by a percentage of the total damage. An extension of this idea is developed later in [17]. The authors in [14] consider different imperfect maintenance actions with random impact on the deterioration level. For authors in [20] the impact of the imperfect maintenance depends on the residual time. Authors in [35, 36] give a large panel of importance measures and other properties for an imperfectly repaired gamma deteriorating system.

In presence of monitoring data, inference procedures, such as the Maximum Likelihood Method are largely used for maintenance and virtual age models. These methods deal with a large panel of data from the complete observation of the maintenance process to censored or scarce data, perfect imperfect or minimal repair, etc. refer to [37, 38, 39, 40, 41, 42, 43]. However, the inference in presence of maintenance and health indicator data for a monitored system is not extensively studied, refer to [44, 45, 46, 47, 48]. The main contribution of this paper is the inference in presence of degradation data and imperfect maintenance dates which is scarcely discussed in the literature.

In a nutshell, the contribution of the paper is as follows:

- Statistical inference on maintenance data for a imperfectly repaired deteriorating system

- Development of virtual degradation distribution function and the related likelihood.
- Proposition of a replacement policy with non constant preventive maintenance cost.

The remainder of this paper is as follows. In Section 2, after the system description, the maintenance model is presented. In Section 3 the inference in presence of data is discussed. Section 4 is devoted to the presentation of a preventive replacement maintenance rule after the model parameters estimation. Eventually, numerical results concerning the preventive policy are given and a conclusion is derived.

2. Problem statement

2.1. System description

The system degradation model is considered to be a gamma process. The author in [49] for the first time models the degradation with a gamma process. This process is monotone increasing which is the behaviour observed in most physical degradation processes. As its paths are discontinuous it can be thought as the accumulation of an infinite number of small shocks. This interpretation gives credence to the model since it is often how the degradation occurs, see [50]. Refer to [51, 52, 8, 53] for more examples using a gamma process to describe a monotonic degradation.

Let be X_t the system degradation level at time t . Let $(X_t)_{t \geq 0}$ be a gamma process. Therefore, $X_0 = 0$ and the on overlapping increments of $(X_t)_{t \geq 0}$ are independent. For all $0 \leq s \leq t$, the increment of $(X_t)_{t \geq 0}$ between s and t , $X_t - X_s$, follows a gamma probability distribution function with shape function $\alpha_1(t) - \alpha_1(s)$, where $\alpha_1(t) = \int_0^t \alpha_1(s) ds > 0$, and scale parameter $\beta_1 > 0$. This probability distribution function can be presented as follows:

$$f_{\alpha_1(t-s), \beta_1}(x) = \frac{1}{\Gamma(\alpha_1(t) - \alpha_1(s)) \beta_1^{\alpha_1(t) - \alpha_1(s)}} x^{\alpha_1(t) - \alpha_1(s) - 1} e^{-\frac{x}{\beta_1}} \mathbf{1}_{\{x \geq 0\}}. \quad (1)$$

The average deterioration speed rate between in $[0, t]$ is $\alpha_1(t) \cdot \beta_1$ and its variance is $\alpha_1(t) \cdot \beta_1^2$. The choice of α_1 and β_2 allows us to model various deterioration behaviours from almost deterministic to very chaotic. Note that the gamma process is a positive process with independent increments, hence it is sensible to use this process to describe the deterioration condition.

2.2. Corrective maintenance rule

A high level of degradation is not acceptable either for economic reasons (quality, high consumption of raw material, poor products, etc...) or for safety reasons (risk of hazardous breakdowns). For instance, one can refer to [54] which gives an example of the problem of maintaining coastal flood barriers. Therefore when the degradation level (health indicator) reaches a given safety level L the system is maintained correctively even if the system is still operating.

Let be T_L first hitting time of a threshold L by a gamma process defined as follows:

$$T_L = \inf\{t > 0, X_t > L, X_0 = 0\} \quad (2)$$

As it is mentioned in [6], T_L has the following cumulative distribution function:

$$\begin{aligned} F_{T_L}(t) &= P(T_L < t) = P(X_t > L - \nu) \\ &= \frac{\Gamma(\alpha_1(t), (L - \nu)\beta_1)}{\Gamma(\alpha_1(t))} = Q(\alpha_1(t), (L - \nu)\beta_1), \end{aligned} \quad (3)$$

where $\Gamma(x, y) = \int_x^\infty z^{y-1} e^{-z} dz$, ($x > 0, y \geq 0$), $Q(x, y) = \frac{\Gamma(x, y)}{\Gamma(x)}$. Therefore, the distribution function of T_F is as follows:

$$f_{T_L}(t) = \frac{\partial F_{T_L}(t)}{\partial t} = \alpha_1'(t) Q'(\alpha_1(t), (L - \nu)\beta_1).$$

Let be T_i the time interval between the $(i-1)$ th and i th maintenance. Let be X_i the deterioration increment between the $(i-1)$ th and i th maintenance. In other words, X_i is the accumulated deterioration in the period beginning just after the $(i-1)$ th maintenance and just before the i th maintenance. At the i th maintenance time, the system deterioration level is decreased of a random amount R_i . After the i th maintenance the system has the virtual degradation $V_i = X_i - R_i$. Let us denote by Y_t the total degradation at time t which is the cumulative deterioration level obtained by counting the degradation and the improvements until time t . For a better comprehension see Fig 1.

In this paper, it is considered that the improvement at the i th corrective maintenance R_i follows a right truncated Gamma distribution denoted by

Figure 1: System of real degradation levels (increments) and life times (duration times) used in this study.

$\Gamma_T(\alpha_2, \beta_2, l, u)$, $\alpha_2 > 0$ and $\beta_2 > 0$, with the characteristic function as follows:

$$\begin{aligned}\Phi_{R_i}(t) = E[e^{itR_i}] &= \frac{1}{F_{\alpha_2, \beta_2}(u) - F_{\alpha_2, \beta_2}(l)} \int_l^u \frac{\beta_2^{\alpha_2} r^{\alpha_2-1} e^{-\beta_2 r} e^{itr}}{\Gamma(\alpha_2)} dr \\ &= \frac{1}{F_{\alpha_2, \beta_2}(u) - F_{\alpha_2, \beta_2}(l)} \left(\frac{\beta_2 - it}{\lambda} \right)^{-\alpha_2} \int_{l \frac{\beta_2 - it}{\beta_2}}^{u \frac{\beta_2 - it}{\beta_2}} \frac{\beta_2^{\alpha_2} z^{\alpha_2-1} e^{-\beta_2 z}}{\Gamma(\alpha_2)} dz, \\ &= \frac{P(it, \alpha_2, \beta_2, l, u)}{P(0, \alpha_2, \beta_2, l, u)} \Phi_X(t).\end{aligned}$$

where $F_{\alpha, \beta}$ is the cumulative distribution function of a gamma distribution with shape parameter α and le parameter λ .

$$P(it, \alpha, \beta, l, u) = F_{\alpha, \beta} \left(\frac{u(\beta - it)}{\beta} \right) - F_{\alpha, \beta} \left(\frac{l(\beta - it)}{\beta} \right). \quad (4)$$

The following Lemma gives the probability distribution function of the deterioration level just before the i th maintenance X_i conditionally to the deterioration level just after the $i - 1$ th maintenance which can be noticed it is independent of the reduction variables R_i .

Lemma 1 *Let be the event $L_i = \{X_i + V_{i-1} > L\}$. The probability distribution function of the deterioration level just before the i th maintenance X_i conditionally to the deterioration level just after the $i - 1$ th maintenance V_i is as follows:*

$$f_{X_i|V_{i-1}}^{L_i}(x) = \frac{\beta_1^{\alpha_1(t_i) - \alpha_1(t_{i-1})} x^{\alpha_1(t_i) - \alpha_1(t_{i-1}) - 1} e^{-\beta_1 x}}{\Gamma(\alpha_1(t_i) - \alpha_1(t_{i-1}), \beta_1(L - \nu_{i-1}))} \mathbf{1}_{x \in \mathbb{R}^+}. \quad (5)$$

For the proof refer to Appendix A

3. Inference

In the model under consideration, there are two classes of parameters: the maintenance and degradation parameters. The maintenance parameter are α_2 and b_2 the parameters of the truncated gamma distribution associated

to the degradation reductions R_i , $i = 1, 2, \dots$. The degradation parameters are the α_1 and b_1 . The statistical inference aims to estimate the degradation parameters and it will help to understand the underlying degradation phenomenon and to predict the future behaviour. The maintenance parameters estimation, permits in presence of corrective maintenance data to learn about an unknown policy carried out on the system and to propose an optimal preventive maintenance policy improving the system's yield and extending its lifetime.

In presence of large size data, the Maximum Likelihood (ML) method is proven to be efficient, refer to [55, 56, 57]. The robustness and efficiency results are all asymptotic and based on independent and identically distributed data, [58, 59, 60, 61]. Under some specific conditions, the asymptotic efficiency of the maximum likelihood method is proved in the case of non-identically distributed data refer to [56, 62, 63]. Moreover, when the parameters of interest require only the maximisation of a conditional likelihood function, authors in [64, 65] among others developed some asymptotic efficiency results. Based on these results, to apply the ML method, the first step is the develop the likelihood function with respect to to available data. In this paper the two following cases are considered:

- Only degradation levels, just before the corrective maintenance are available.
- Only maintenance time intervals are available.

For each case, the likelihood function is developed and a numerical example is given.

3.1. Inference with degradation data

Let's consider the following sample of deterioration level just before corrective maintenance times $x_1, x_2, \dots, x_n$. In order to estimate the parameters of the degradation and the maintenance, the likelihood of $X_1, X_2, \dots, X_n$ should be derived. The first step is to derive the likelihood of $X_1, X_2, \dots, X_n$ conditional to the deterioration levels just before corrective maintenance times $V_1, V_2, \dots, V_n$. This conditional likelihood is derived in the next proposition

Proposition 1 *Let be $X_1, X_2, \dots, X_n$ the degradation levels just before corrective maintenance times and $V_1, V_2, \dots, V_n$ the virtual degradation just after*

corrective maintenance. The likelihood can be written as follows.

$$\begin{aligned}
L(X_1, X_2, \dots, X_n | V_1, V_2, \dots, V_{n-1}) &= \prod_{k=2}^n f_{X_k|V_{k-1}}^{L_k}(x_k) \cdot f_{-R_k}(v - z) \cdot f_{X_1}(x_1) \\
&= \prod_{k=2}^n \frac{\beta_1^{\alpha_1(t_k) - \alpha_1(t_{k-1})} x_k^{\alpha_1(t_k) - \alpha_1(t_{k-1}) - 1} e^{-\beta_1 x_k}}{\Gamma(\alpha_1(t_k) - \alpha_1(t_{k-1}), \beta_1(L - \nu_{k-1}))}. \tag{6}
\end{aligned}$$

$$\frac{\beta_1^{\alpha_1(t_1)} x_1^{\alpha_1(t_1) - 1} e^{-\beta_1 x_1}}{\Gamma(\alpha_1(t_1))} \times \prod_{k=2}^n \frac{1}{2\pi} \int_{\mathbb{R}} \frac{e^{-is(v-z)} P(-is, \alpha_2(t_k), \beta_2, l, u)}{P(0, \alpha_2(t_k), \beta_2, l, u)} \left(\frac{\beta_2 + is}{\beta_2} \right)^{-\alpha_2(t_k)} dt$$

where the function P is defined in equation (4).

The proof is given in Appendix B.

The likelihood of $X_1, X_2, \dots, X_n$ the degradation levels just before corrective maintenance times is given in Proposition (2).

Proposition 2

$$\begin{aligned}
L(X_1, X_2, \dots, X_n) &= \int \dots \int \left(\prod_{k=2}^n f_{X_k|V_{k-1}}^{L_k}(x_k) \cdot f_{-R_k}(V_k - V_{k-1} - x_k) \cdot f_{X_1}(x_1) \right) \\
&\times \prod_{k=1}^n \frac{1}{2\pi} \int_R e^{-it(V_k - V_{k-1})} \frac{P(i(-t), \alpha_2(t_k), \beta_2, l, u)}{P(0, \alpha_2(t_k), \beta_2, l, u)} \left[\frac{\beta_1 - it}{\beta_1} \right]^{-\alpha_2(t_k)} \left[\frac{\beta_2 - i(-t)}{\beta_2} \right]^{-\alpha_2(t_k)} dt dV_1 dV_2 \dots dV_n \\
&= \int \dots \int \prod_{k=2}^n \left[\frac{\beta_1^{\alpha_1(t_k) - \alpha_1(t_{k-1})} x_k^{\alpha_1(t_k) - \alpha_1(t_{k-1}) - 1} e^{-\beta_1 x_k}}{\Gamma(\alpha_1(t_k) - \alpha_1(t_{k-1}), \beta_1(L - V_{k-1}))} \right] \cdot \frac{\beta_1^{\alpha_1(t_1)} x_1^{\alpha_1(t_1) - 1} e^{-\beta_1 x_1}}{\Gamma(\alpha_1(t_1))} \\
&\cdot \prod_{k=2}^n \frac{1}{2\pi} \int_R e^{-is(V_k - V_{k-1} - x_k)} \frac{P(-is, \alpha_2(t_k), \beta_2, l, u)}{P(0, \alpha_2(t_k), \beta_2, l, u)} \left(\frac{\beta_2 - i(-s)}{\beta_2} \right)^{-\alpha_2(t_k)} ds \\
&\times \prod_{k=1}^n \frac{1}{2\pi} \int_R e^{-it(V_k - V_{k-1})} \frac{P(i(-t), \alpha_2(t_k), \beta_2, l, u)}{P(0, \alpha_2(t_k), \beta_2, l, u)} \left(\frac{\beta_2 - it}{\beta_2} \right)^{-\alpha_1(t_k)} \left(\frac{\beta_2 - i(-t)}{\beta_2} \right)^{-\alpha_2(t_k)} dt dV_1 dV_2 \dots dV_n \tag{7}
\end{aligned}$$

where P is defined in equation (4).

The proof is given in Appendix C.

Let's consider the power law function as the shape function $\alpha_2(t) = a_2 t^{b_2}$ and $\alpha_1(t) = a_1 t^{b_1}$. The profile likelihood corresponding to the likelihood function given in equation (7) is depicted in Figures 2 and 3. For given parameters values, the maximum profile likelihood in function of can be obtained.

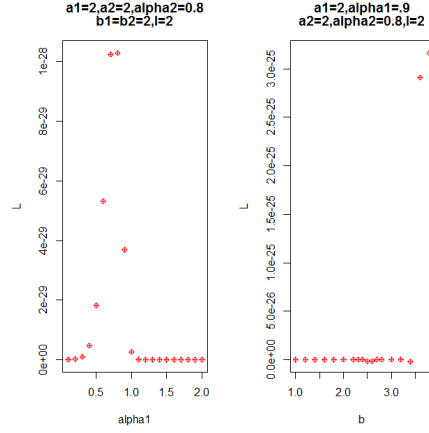


Figure 2: Likelihood function of equation (7) respectively in terms of α_1 and β ($\beta_1=\beta_2 = \beta$) with different values of parameters

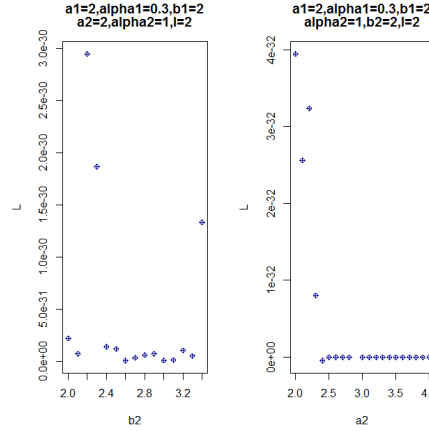


Figure 3: Likelihood function of equation (7) for homogeneous case of repair efficiency parameters ($a_2 = 1$) respectively in terms of b_2 and a_2 with different values of parameters

3.2. Inference in presence of maintenance cycle times data

Let be $T_1, T_2, \dots, T_n$ the time duration of a maintenance cycle. If only this information is available or if it is decided to estimate parameters only based on this data the likelihood can be derived as in Proposition 3.

Proposition 3 *Let be $T_1, T_2, \dots, T_n$ the time duration of a maintenance cycle. The likelihood is as follows:*

$$L^*(T_1, T_2, \dots, T_n) = \int_0^L \dots \int_0^L \int_0^\infty \dots \int_0^\infty \prod_{i=1}^n f_{T_i|V_{i-1}^*}^*(t_i) f_{V_{n-1}^*}(\nu_{n-1}^*) \\ f_{V_{n-2}^*}(\nu_{n-2}^*) \dots f_{V_2^*}(\nu_2^*) f_{V_1^*}(\nu_1^*) dV_0^* \dots dV_{n-1}^* dV_{n-1} dV_{n-2} \dots dV_0 \quad (8)$$

The proof is given in Appendix D.

The likelihood of $T_1, T_2, \dots, T_n$ depends on the probability distribution function of the virtual age at maintenance times. To derive the distribution of the virtual age at the i th maintenance time, the latter should be properly defined. It is considered that V_i^* is the hitting time of L at after the $(i - 1)$ th maintenance. Two probability distribution functions V_i^* are considered: truncated gamma distribution for an increment between repair and failure and a gamma distribution. The probability density function associated to these two cases are developed in Appendix E.

An example of profile likelihood associated to equation (8) when $\alpha_2(t) = a_2 t^{b_2}$ and $\alpha_1(t) = a_1 t^{b_1}$.

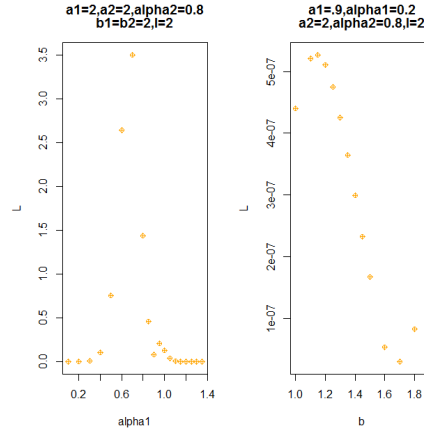


Figure 4: The profile likelihood function of equation (8) respectively in terms of α_1 and b ($b_1=b_2 = b$) with different values of parameters

4. Replacement planning and numerical examples

Once the degradation and maintenance model parameters are estimated, based on this estimation a replacement policy can be planned. After n imperfect corrective maintenance, a replacement is carried out. The idea is

to examine the impact of the variation of n and the total long run average cost. The maintenance parameter to optimise is n , the number of imperfect maintenance actions before replacements. The following procedure is carried out

- In presence of data, estimate the degradation and maintenance parameters
- According to the parameters estimate propose an optimal n , number of imperfect maintenance actions before the replacement

The unit cost related to this policy are as follows. The unit cost C_p induced at each imperfect corrective repair. The unit cost C_c induced at each corrective replacement. Let be $C(t)$ the cost at time t . $C(t) = C_p \sum_{i=1}^{\infty} \mathbb{1}_{\{T_i < t\}} + (C_c - C_p) \sum_{i=1}^{\infty} \mathbb{1}_{\{T_{in} < t\}}$

Let be T a replacement cycle. The total cost during a replacement cycle T given as follows: $C(T) = nC_p + C_c$. Since the system is replaced after n preventive, by the renewal theory, the long run average cost is given as follows:

$$C_{\infty}(n) = \lim_{t \rightarrow \infty} \frac{C_t}{t} = \frac{\mathbb{E}(C(T))}{\mathbb{E}(T)} = \frac{C(T)}{\mathbb{E}(T)} = \frac{(n-1)C_p + C_c}{\mathbb{E}(T)}.$$

As mentioned in [66, 67] to find an optimal n , the following inequalities are solved $C_{\infty}(n) \leq C_{\infty}(n+1)$ and $C_{\infty}(n) < C_{\infty}(n-1)$. Since $T = \sum_{i=1}^n T_i$, for $n > 0$, the inequalities imply,

$$\frac{\mathbb{E}(\sum_{i=1}^n T_i) - (n-1)\mathbb{E}(T_{n+1})}{\mathbb{E}(T_{n+1})} \geq \frac{C_c}{C_p} \quad \text{and} \quad \frac{\mathbb{E}(\sum_{i=1}^{n-1} T_i) - (n-2)\mathbb{E}(T_n)}{\mathbb{E}(T_n)} < \frac{C_c}{C_p}.$$

To derive the cost let us recall that $\mathbb{E}(T) = \int_0^{+\infty} \mathbb{P}(T > t) dt$ and

$$\mathbb{P}(T \leq t) = \mathbb{P}\left(\sum_{i=1}^n T_i \leq t\right). \quad (9)$$

This distribution can be calculated by an n fold convolution using cumulative distribution function of T_i . The distribution function of T_i can be obtained by the product of the conditional distribution of T_i to V_i^* defined in equation (D.2) and the distribution of V_i^* . The distribution of V_i^* is subject to

Case	b_2	β_2
1	1	1.5
2	1.5	1
3	1	1,5
4	1. 5	2

Table 1: The table of different numerical cases.

the definition of V_i^* . Moreover, as n grows the n fold convolution become burdensome, that is why for numerical optimisation the Monte Carlo simulations are considered. In Figure 5, an example of the empirical distribution of T_1, T_2, T_3 and T_4 are depicted.

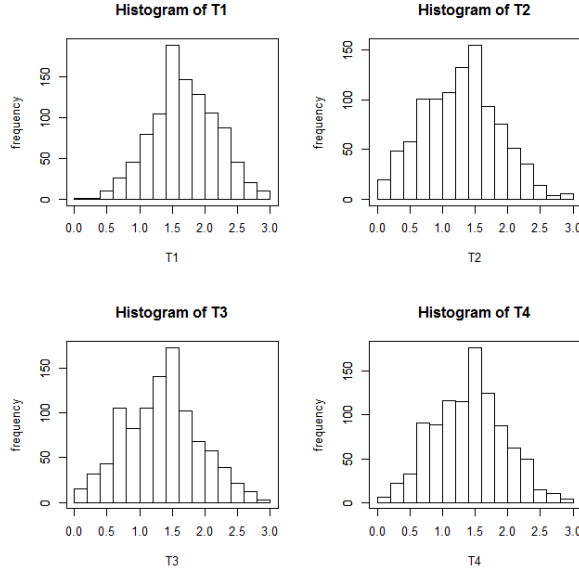


Figure 5: The empirical distribution of T_1, T_2, T_3 and T_4 for parameters $\alpha_1 = 1.5, \beta_1 = 2, \alpha_2 = 2$.

For $n \in \{4, 5, 10\}$ the empirical distribution of $TT_n = \sum_{i=1}^n T_i$ is depicted in Figure 6. In Figures 7 and 8 the cost evolution in function of n of for cases presented in Table 1 for $C_c \in \{70, 100\}$ and $C_p \in \{40, 100\}$ is depicted. It can be noticed that there is no global optimum and the replacement can be carried out arbitrary after a large number of times.

Since a constant C_p does not lead to an optimal cost. To study the

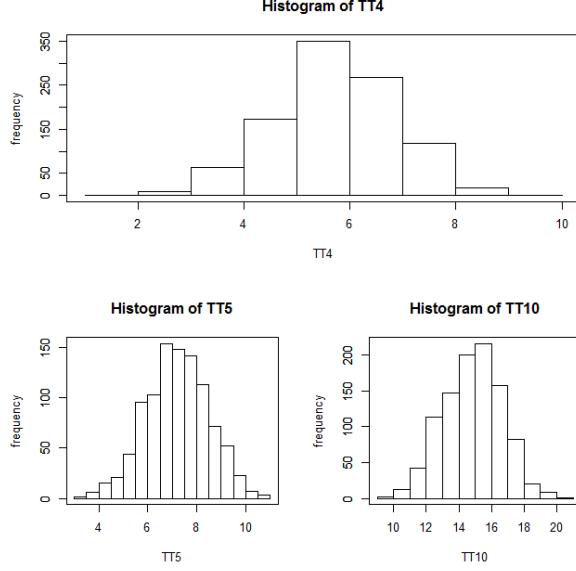


Figure 6: The empirical distribution of TT_4 , TT_5 and TT_{10} for parameters $\alpha_1 = 1.5$, $\beta_1 = 2$, $\alpha_2 = 2$.

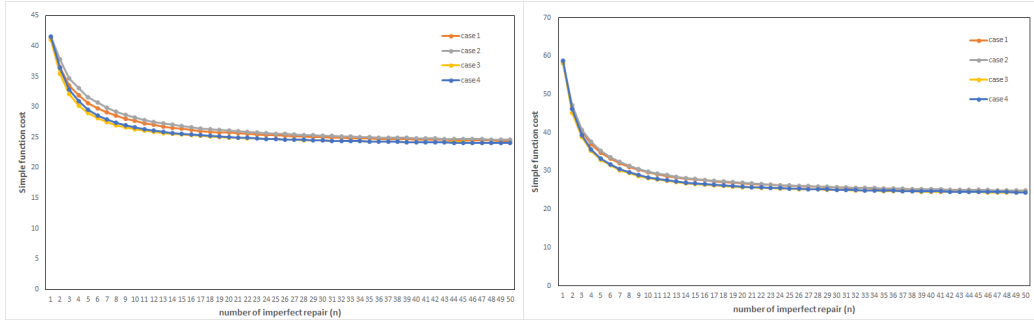


Figure 7: The cost evolution in function of n with a constant $C_p = 40$ and $C_c \in \{70, 100\}$, for parameters $\alpha_1 = 1.5$, $\beta_1 = 2$, $\alpha_2 = 2$ and four cases presented in Table 1.

impact of time and the number of imperfect repairs on the preventive cost the following cost cases are considered.

- $C_p(k) = s + k\delta$, with $s, \delta > 0$, k is the number of imperfect repair.
- $C_p(k) = s + k^2\delta$, with $s, \delta > 0$, k is the number of imperfect repair.

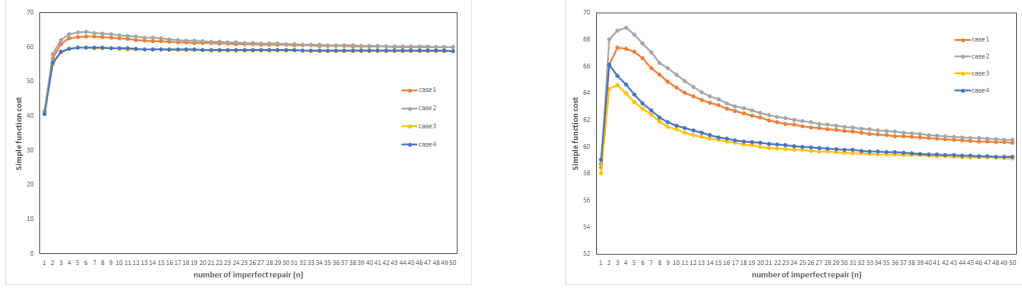


Figure 8: The cost evolution in function of n with a constant $C_p = 100$ and $C_c \in \{70, 100\}$, for parameters $\alpha_1 = 1.5$, $\beta_1 = 2$, $\alpha_2 = 2$ and four parameter cases presented in Table 1.

- $C_p(t) = \delta \exp(\lambda t)$, with $\lambda, \delta > 0$, t , is the cumulative time until the last repair (tti).
- $C_p(k, t) = \delta(\exp(\lambda t) + k)$, with $\lambda, \delta > 0$, k is the number of imperfect repair and t is the cumulative time until the last repair(TT_n).
- $C_p(t) = s(\Psi(t, \mu, \sigma)) + \delta$ with $s, \delta > 0$, Ψ is the cumulative Normal distribution (I wrote some theory for this part).
- $C_p(k) = s(pbinom(k, n + 15, p)) + \delta$ with $s, \delta > 0$, $pbinom$ is the cumulative Binomial distribution (I wrote some theory for this part).

Figures 9 and 10 give the cost evolution for $C_c = 100$ with 1000 repetitions for a non constant preventive cost. As it can be noticed, by a non constant preventive cost the replacement policy can be optimised.

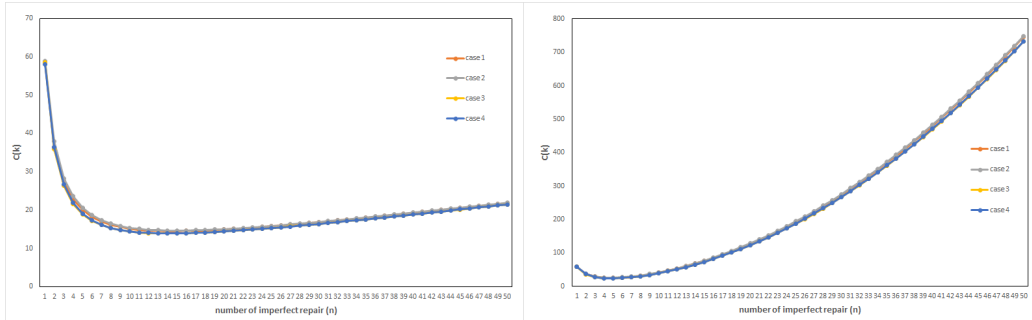


Figure 9: The cost evolution in function of for linear and quadratic C_p , respectively as $C_p(k) = s + k\delta$ (left) and $C_p(k) = s + k^2\delta$ (right) with $C_c = 100$ for parameters $\alpha_1 = 1.5$, $\beta_1 = 2$, $\alpha_2 = 2$ and four parameter cases presented in Table 1.

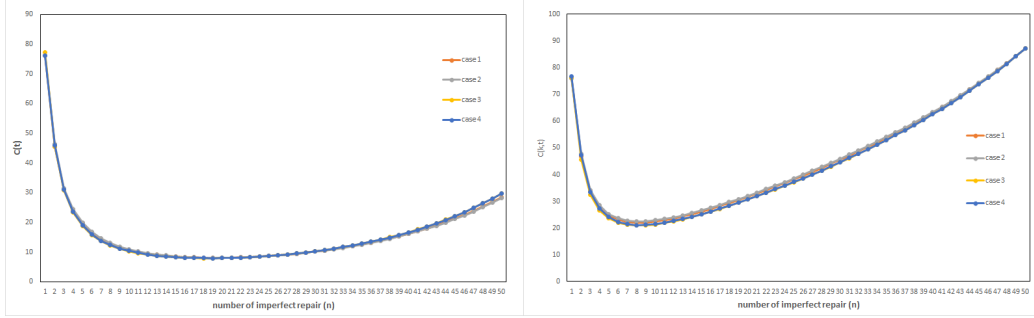


Figure 10: The cost evolution in exponential function of t and (t, k) for C_p , respectively as $C_p(t) = \delta \exp(\lambda t)$ (left) and $C_p(k, t) = \delta(\exp(\lambda t) + k)$ (right) with $C_c = 100$ for parameters $\alpha_1 = 1.5$, $\beta_1 = 2$, $\alpha_2 = 2$ and four parameter cases presented in Table 1.

5. Conclusion

This paper considers a deteriorating system modelled by a gamma process. First the statistical inference for imperfect correctively maintained system is considered. The maintenance efficiency is random following a truncate gamma distribution. The likelihood of degradation levels at failure time and the likelihood of failure times are derived. The parameter estimates are obtained by numerical optimisation. Based on estimated data a replacement policy can be proposed. The policy proposes to replace the system after a given number of imperfect repairs. The average maintenance cost evaluation is detailed and sensitivity analyses to different parameters and costs are carried out. As short term perspective, one can focus on the maximum likelihood estimates performance analysis. The notion of virtual degradation can be more explored in the future.

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Appendix A. Proof of Lemma 1

Let's consider the i failure event as follows: $L_i = \{X_i + V_{i-1} > L\}$. The cumulative distribution function of X_i subject to L_i and the virtual degradation V_i is as follows:

$$\begin{aligned} F_{X_i|V_{i-1}}^{L_i}(x) &= \mathbb{P}(X_i < x | V_{i-1} = \nu_{i-1}, X_i + V_{i-1} > L) \\ &= \frac{\mathbb{P}(L - \nu_{i-1} < X_i < x | V_{i-1} = \nu_{i-1})}{\mathbb{P}(X_i > L - \nu_{i-1} | V_{i-1} = \nu_{i-1})}. \end{aligned} \quad (\text{A.1})$$

In order to obtain distribution function in equation (A.1), two choices can be made: consider the virtual degradation as observed values (ν_i) or as random variables (v_i), which in the first case we can write the likelihood function in form of joint distribution of X_i s and ν_i s and in the second case one can obtain the conditional likelihood considered later in the joint distribution of v_i s. Let's denote $t_i = t_{V_i}$, hence

$$X_i | V_{i-1} \sim \Gamma(\alpha_1(t_{V_{i-1}}), \beta_1) \equiv \Gamma(\alpha_1(t_i), \beta_1)$$

therefore

$$f_{X_i|V_{i-1}}(x) = \frac{\beta_1^{\alpha_1(t_i)} x^{\alpha_1(t_i)-1} e^{-\beta_1 x}}{\Gamma(\alpha_1(t_i))}, \quad (\text{A.2})$$

By putting this formula on Eq. (A.1) we have

$$\begin{aligned}
F_{X_i|V_{i-1}}^{L_i}(x) &= \frac{\int_{L-\nu_{i-1}}^x \frac{\beta_1^{\alpha_1(t_i)} z^{\alpha_1(t_i)-1} e^{-\beta_1 z}}{\Gamma(\alpha_1(t_i))} dz}{\int_L^\infty \frac{\beta_1^{\alpha_1(t_i)} z^{\alpha_1(t_i)-1} e^{-\beta_1 z}}{\Gamma(\alpha_1(t_i))} dz} \\
&= \frac{\gamma(\alpha_1(t_i), \beta_1 x) - \gamma(\alpha_1(t_i), \beta_1(L - \nu_{i-1}))}{\Gamma(\alpha_1(t_i), \beta_1(L - \nu_{i-1}))}.
\end{aligned}$$

The probability distribution function is as follows,

$$\begin{aligned}
f_{X_i|V_{i-1}}^{L_i}(x) &= \frac{\partial F_{X_i|V_{i-1}}^{L_i}(x)}{\partial x} = \frac{\beta_1^{\alpha_1(t_i)} x^{\alpha_1(t_i)-1} e^{-\beta_1 x}}{\Gamma(\alpha_1(t_i), \beta_1(L - \nu_{i-1}))} \\
&= \frac{\beta_1^{\alpha_1(t_i)-\alpha_1(t_{i-1})} x^{\alpha_1(t_i)-\alpha_1(t_{i-1})-1} e^{-\beta_1 x}}{\Gamma(\alpha_1(t_i) - \alpha_1(t_{i-1}), \beta_1(L - \nu_{i-1}))}. \tag{A.3}
\end{aligned}$$

Appendix B. Proof Proposition 1

$$L(X_1, X_2, \dots, X_n, V_1, V_2, \dots, V_{n-1}) = \prod_{i=2}^n p(X_i, V_{i-1} \mid X_{i-1}, V_{i-2}) f_{x_1}(x)$$

For simplification let's introduce the following notations:

$$A_i = (X_i, V_{i-1}), \quad B_i = A_i \mid A_{i-1} \tag{B.1}$$

$$f_{B_i} = f_{A_i|A_{i-1}} = \frac{f_{A_i, A_{i-1}}}{f_{A_{i-1}}}, \quad f_{B_i|L_i} = f_{[A_i|A_{i-1}]|L_i} = \frac{f_{(A_i, A_{i-1})|L_i}}{f_{A_{i-1}|L_i}} \tag{B.2}$$

$$\begin{aligned}
f_{(A_i, A_{i-1})|L_i} &= f_{X_i, V_{i-1}, X_{i-1}, V_{i-2}|L_i}(x, \nu_1, y, \nu_2) \\
&= f_{A_i|A_{i-1}, L_i} \cdot f_{A_{i-1}|L_i} \\
&= f_{X_i, V_{i-1}|X_{i-1}, V_{i-2}, L_i}(x, \nu_1) \cdot f_{X_{i-1}, V_{i-2}|L_i}(y, \nu_2) \\
&= f_{X_i|V_{i-1}, X_{i-1}, V_{i-2}, L_i}(x) \cdot f_{V_{i-1}|X_{i-1}, V_{i-2}, L_i}(\nu_1) \cdot f_{X_{i-1}, V_{i-2}|L_i}(y, \nu_2) \\
&= f_{X_i|V_{i-1}}^{L_i}(x) \cdot f_{V_{i-1}|X_{i-1}, V_{i-2}, L_i}(\nu_1) \cdot f_{X_{i-1}, V_{i-2}|L_i}(y, \nu_2) \tag{B.3}
\end{aligned}$$

The function $f_{X_i|V_{i-1}}^{L_i}$ is given in equation (A.1). Since

$$\begin{aligned}
F_{V_i|V_{i-1}, X_i}(v) &= \mathbb{P}(V_i < v | V_{i-1}, X_i) = \mathbb{P}(V_{i-1} + X_i - R_i < v | V_{i-1}, X_i) \\
&= \mathbb{P}(-R_i < v - V_{i-1} - X_i | V_{i-1}, X_i) \\
&= 1 - P(R_i < -v + z | V_{i-1}, X_i, V_{i-1} + X_i = z) \\
&= 1 - F_{R_i}(-v + z)
\end{aligned}$$

and $R_i \sim \Gamma(\alpha_2(t_i), \beta_2, l, u)$. Therefore,

$$f_{V_i|V_{i-1}, X_i}(v) = f_{R_i}(-v + z)$$

Calculation of $f_{X_i, V_{i-1}|X_{i-1}, V_{i-2}}$.

$$\begin{aligned}
f_{X_i, V_{i-1}|X_{i-1}, V_{i-2}, L_i}(x, \nu_1) &= f_{X_i|V_{i-1}, X_{i-1}, V_{i-2}, L_i}(x) \cdot f_{V_{i-1}|X_{i-1}, V_{i-2}, L_i}(\nu_1) \cdot f_{X_{i-1}, V_{i-2}|L_i}(y, \nu_2) \\
&= f_{X_i|V_{i-1}}^{L_i}(x) \cdot f_{-R_i}(r)
\end{aligned}$$

where

$$\begin{aligned}
f_{X_{i-1}, V_{i-2}|L_i}(y, \nu_2) &= f_{A_{i-1}|L_i} \\
&= f_{X_{i-1}|V_{i-2}, L_i}(y) \cdot f_{V_{i-2}|L_i}(\nu_2) \\
&= f_{X_{i-1}|V_{i-2}}(y) \cdot f_{V_{i-2}}(\nu_2)
\end{aligned}$$

$$\begin{aligned}
f_{B_i|L_i} &= \frac{f_{(A_i, A_{i-1})|L_i}}{f_{A_{i-1}|L_i}} = \frac{f_{X_i, V_{i-1}|X_{i-1}, V_{i-2}, L_i}(x, \nu_1) \cdot f_{X_{i-1}|V_{i-2}, L_i}(y)}{f_{X_{i-1}|V_{i-2}, L_i}(y)} \\
&= f_{X_i|V_{i-1}}^{L_i}(x) \cdot f_{-R_i}(v - z)
\end{aligned}$$

Appendix C. Proof proposition 2

$$L(X_1, \dots, X_n) = \int \dots \int L(X_1, \dots, X_n, V_1, \dots, V_{n-1}) \cdot h(V_1, \dots, V_{n-1}) dV_1 \dots dV_{n-1} \quad (C.1)$$

$$h(V_1, V_2, \dots, V_{n-1}) = f_{V_{n-1}|V_{n-2}, \dots, V_1}(v_{n-1}) \cdots f_{V_2|V_1}(v_2) f_{V_1}(v_1) \quad (\text{C.2})$$

Let's denote $W_k = X_k - R_k$. Since

$$V_k = V_{k-1} + (X_k - R_k) = V_{k-1} + W_k, \quad (\text{C.3})$$

$$\begin{aligned} F_{V_k}(V_k | V_{k-1} = \nu_{k-1}) &= P(V_k < \nu_k | V_{k-1} = \nu_{k-1}) \\ &= P(V_{k-1} + X_k - R_k < \nu_k | V_{k-1} \\ &= \nu_{k-1}) = P(X_{k-1} - R_{k-1} < \nu_k - \nu_{k-1}) \quad (\text{C.4}) \\ &= F_{W_k}(\nu_k - \nu_{k-1}), \quad (\text{C.5}) \end{aligned}$$

Let's recall that $X_j \sim \Gamma(\alpha_1(t_j), \beta_1)$ and $R_i \sim \Gamma_T(\alpha_2, \beta_2, l, u)$, the following notations are used.

$$X \sim \Gamma(\alpha, \beta), \quad F_{\alpha, \beta}(s) = \int_0^s \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} dx, \quad \Phi_X(t) = E[e^{itx}] = \left(\frac{\beta - it}{\beta}\right)^{-\alpha},$$

Therefore, based on these notations,

$$\begin{aligned} \Phi_{W_k}(t) &= \Phi_{X_k - R_k}(t) = \Phi_{X_k}(t) \Phi_{R_k}(-t) \quad (\text{C.6}) \\ &= \left[\frac{P(i(-t), \alpha_2(t_k), \beta_2, l, u)}{P(0, \alpha_2(t_k), \beta_2, l, u)} \right] \left[\frac{\beta_1 - it}{\beta_1} \right]^{-\alpha_1(t_k)} \left[\frac{\beta_2 - i(-t)}{\beta_2} \right]^{-\alpha_2(t_k)} \end{aligned}$$

Which implies that the probability distribution of W_k can be derived as follows:

$$f_{W_k}(w) = F'_{W_k}(w) = \frac{1}{2\pi} \int_R e^{-itw} \Phi_{W_k}(t) dt, \quad (\text{C.7})$$

From equations (C.4) and (C.7) one can derive the probability distribution function of $V_k | V_{k-1}$ as follows:

$$\begin{aligned} f(\nu_k | \nu_{k-1}) &= \frac{1}{2\pi} \int_R e^{-it(\nu_k - \nu_{k-1})} \left(\frac{P(i(-t), \alpha_2(t_j), \beta_2, l, u)}{P(0, \alpha_2(t_j), \beta_2, l, u)} \right) \\ &\quad \times \left(\frac{\beta_1 - it}{\beta_1} \right)^{-\alpha_1(s)} \left(\frac{\beta_2 - i(-t)}{\beta_2} \right)^{-\alpha_2(t_j)} dt \quad (\text{C.8}) \end{aligned}$$

The calculation of the likelihood is straightforward. $\diamond$

Appendix D. Proof of Porposition 3

The likelihood of $T_1, T_2, \dots, T_n$ can be derived by using the conditional joint distribution of $T_1, T_2, \dots, T_n$ subject to $V_1^*, \dots, V_{n-1}^*$ and the joint distribution of the virtual ages $V_1^*, \dots, V_{n-1}^*$.

The conditional joint distribution of $T_1, T_2, \dots, T_n$ subject to $V_1^, \dots, V_{n-1}^*$.* This conditional joint distribution can be expressed as follows:

$$\begin{aligned}
L^*(T_1, T_2, \dots, T_n | V_1^*, \dots, V_{n-1}^*) &= P(T_n | t_1, t_2, \dots, t_{n-1}, V_1^*, \dots, V_{n-1}^*) \times P(T_1, T_2, \dots, T_{n-1} | V_1^*, \dots, V_{n-1}^*) \\
&= f_{T_n | \nu_{n-1}^*}^*(t_n) P(T_{n-1} | t_1, t_2, \dots, t_{n-2}, V_1^*, \dots, V_{n-1}^*) P(T_1, T_2, \dots, T_{n-2} | V_1^*, \dots, V_{n-1}^*) \\
&= \prod_{i=1}^n f_{T_i | V_{i-1}^*}^*(t_i). \tag{D.1}
\end{aligned}$$

The probability distribution function of T_i can be derived as follows,

$$\begin{aligned}
F_{T_i | V_{i-1}^*}(t_i) &= \frac{F_{T_L}(t_i + V_{i-1}^*) - F_{T_L}(\nu_{i-1}^*)}{1 - F_{T_L}(V_{i-1}^*)} \\
&= \frac{Q(\alpha_1(t_i + V_{i-1}^*), (L - V_{i-1})\beta_1) - Q(\alpha(V_{i-1}^*), (L - V_{i-1})\beta_1)}{1 - Q(\alpha_1(V_{i-1}^*), (L - V_{i-1})\beta_1)},
\end{aligned}$$

therefore the distribution function of T_i is as follows:

$$f_{T_i | V_{i-1}^*}(t_i) = \frac{\partial F_{T_i | V_{i-1}^*}(t_i)}{\partial t_i} = \frac{\frac{\partial}{\partial t_i} [Q(\alpha_1(t_i + V_{i-1}^*), (L - V_{i-1})\beta_1)]}{1 - Q(\alpha_1(V_{i-1}^*), (L - V_{i-1})\beta_1)},$$

which can be derived by

$$\begin{aligned}
\frac{\partial}{\partial t_i} [Q(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1)] &= \frac{\partial}{\partial t_i} \left(\frac{\Gamma(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1)}{\Gamma(\alpha_1(t_i + \nu_{i-1}^*))} \right) \\
&= \frac{\left(\frac{\partial}{\partial t_i} \alpha(t_i + \nu_{i-1}^*) \right)}{\Gamma(\alpha_1(t_i + \nu_{i-1}^*))} \\
&\times \{ [\Gamma(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1) \ln((L - \nu_{i-1})\beta_1) \\
&+ G_{\frac{3}{2} \frac{0}{3}} \left(\begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \middle| \alpha(t_i + \nu_{i-1}^*) \right) (L - \nu_{i-1})\beta_1 \\
&- \Gamma(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1) \Psi(\alpha_1(t_i + \nu_{i-1}^*)) \} \tag{D.2}
\end{aligned}$$

where G is the Meijer function defined as follows,

$$G_{\frac{3}{2}}^{\frac{3}{2}} \left(\begin{matrix} 1 & 1 \\ 0 & 0 \end{matrix} \middle| a \right) z = \Gamma(a)[\Psi(a) - \ln(z)] + \left(\frac{z^a}{a^2}\right) {}_2F_2(a, a; a+1, a+1; -z)$$

and ${}_pF_q$, $p > 0$ and $q > 0$, is the generalized hypergeometric function, refer to [68]. Therefore,

$$\begin{aligned} \prod_{i=1}^n f_{T_i|\nu_{i-1}^*}(t_i) &= \left(\frac{\partial}{\partial t_i} \alpha_1(t_i + \nu_{i-1}^*)\right) \cdot \frac{\Gamma(\alpha_1(\nu_{i-1}^*))}{\Gamma(\alpha_1(t_i + \nu_{i-1}^*)) \cdot \Gamma(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1)} \\ &\times \{[\ln((L - \nu_{i-1})\beta_1) - \Psi(\alpha_1(t_i + \nu_{i-1}^*))][\Gamma(\alpha_1(t_i + \nu_{i-1}^*), (L - \nu_{i-1})\beta_1) - \Gamma(\alpha_1(t_i + \nu_{i-1}^*))]\} \\ &+ \frac{[(L - \nu_{i-1})\beta_1]^{\alpha_1(t_i + \nu_{i-1}^*)}}{\alpha_1(t_i + \nu_{i-1}^*)^2} \\ &{}_2F_2(\alpha_1(t_i + \nu_{i-1}^*), \alpha_1(t_i + \nu_{i-1}^*); \alpha_1(t_i + \nu_{i-1}^*) + 1, \alpha_1(t_i + \nu_{i-1}^*) + 1; -(L - \nu_{i-1})\beta_1) \} \end{aligned} \quad (D.3)$$

The joint distribution of $V_1^*, V_2^*, \dots, V_{n-1}^*$. Let be $k(V_1^*, V_2^*, \dots, V_{n-1}^*)$ the joint distribution of $V_1^*, V_2^*, \dots, V_{n-1}^*$. It can be noticed that

$$\begin{aligned} k(V_1^*, V_2^*, \dots, V_{n-1}^*) &= f_{V_{n-1}^*|V_{n-2}^*, \dots, V_1^*}(v_{n-1}^*) f_{V_{n-2}^*|V_{n-3}^*, \dots, V_1^*}(v_{n-2}^*) \\ &\cdots f_{V_2^*|V_1^*}(v_2^*) f_{V_1^*}(v_1^*) \\ &= f_{V_{n-1}^*}(v_{n-1}^*) f_{V_{n-2}^*}(v_{n-2}^*) \cdots f_{V_2^*}(v_2^*) f_{V_1^*}(v_1^*). \end{aligned} \quad (D.4)$$

The last equality is the result of the independence of the virtual ages $V_1^*, \dots, V_{n-1}^*$.

Appendix E. Distribution for each cases

Case 1: Truncated gamma distrubition In this case the degradation level after $i - 1$ th maintenance will not exceed $x_i + v_{i-1}$. Therefore, the degraation increment during $[0, t]$, where $t < t_i$ follows a truncated gamma distrubution as follows: $X_t \sim \Gamma(\alpha_1(t), \beta_1, 0, x_i + \nu_{i-1})$

$$\begin{aligned}
F_{V_i^*}(t) &= P(V_i^* < t) = P(X_t > \nu_i) \\
&= \frac{1}{F(x_i + \nu_{i-1}, \alpha_1(t), \beta_1) - F(0, \alpha_1(t), \beta_1)} \int_{\nu_i}^{\infty} \frac{\beta_1^{\alpha_1(t)} x^{\alpha_1(t)-1} e^{-b_1 x}}{\Gamma(\alpha_1(t))} dx \\
&= \frac{1}{F(x_i + \nu_{i-1}, \alpha_1(t), \beta_1)} \int_{b\nu_i}^{\infty} \frac{y^{\alpha_1(t)-1} e^{-y}}{\Gamma(\alpha_1(t))} dy \\
&= \frac{1}{F(x_i + \nu_{i-1}, \alpha_1(t), \beta_1)} \frac{\Gamma(\alpha_1(t), b_1 \nu_i)}{\Gamma(\alpha_1(t))} \\
&= \frac{1}{F(x_i + \nu_{i-1}, \alpha_1(t), \beta_1)} Q(\alpha_1(t), \beta_1 \nu_i)
\end{aligned}$$

The probability distribution of V_i^* is derived as follows:

$$\begin{aligned}
f_{V_i^*}(t) &= \frac{\partial P(X_t > \nu_i)}{\partial t} = \frac{\partial [\frac{Q(\alpha_1(t), \beta_1 \nu_i)}{F(x_i, \alpha_1(t), \beta_1)}]}{\partial t} \\
&= \frac{\partial}{\partial t} \left[\frac{\Gamma(\alpha_1(t), \beta_1 \nu_i)}{\gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))} \right] = \frac{\partial}{\partial t} \left[\frac{\Gamma(\alpha_1(t), \beta_1 \nu_i)}{\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))} \right] = \frac{\partial}{\partial t} \alpha_1(t) \\
&\times \frac{\Gamma(\alpha_1(t), \beta_1 \nu_i) \log(b\nu_i) + \Gamma(\alpha_1(t)) [\Psi(\alpha_1(t)) - \log(\beta_1 \nu_i)]}{\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))} \\
&\times \frac{\frac{(b_1 \nu_i)^{\alpha_1(t)}}{(\alpha_1(t))^2} {}_2F_2(\alpha_1(t), \alpha_1(t); \alpha_1(t) + 1, \alpha_1(t) + 1; -\beta_1 \nu_i)}{\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))} \\
&- \frac{\{\log(\beta_1(x_i + \nu_{i-1})) [\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))]\}}{(\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1})))^2} \\
&- \frac{\frac{(\beta_1(x_i + \nu_{i-1}))^{\alpha_1(t)}}{(\alpha_1(t))^2} {}_2F_2(\alpha_1(t), \alpha_1(t); \alpha_1(t) + 1, \alpha_1(t) + 1; -\beta_1(x_i + \nu_{i-1})) \Gamma(\alpha_1(t), \beta_1 \nu_i)}{(\Gamma(\alpha_1(t)) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1})))^2}
\end{aligned} \tag{E.1}$$

where γ is the incomplete gamma function.

Case 2: gamma distribution In this case, the visual age and the deterioration level can be linked as follows:

$$\begin{aligned}
P(V_i^* \leq t) &= P(X_t > \nu_i, X_t < x_i + \nu_{i-1}) = \int_{\nu_i}^{x_i + \nu_{i-1}} \frac{\beta_1^{\alpha_1(t)} x^{\alpha_1(t)-1} e^{-\beta_1 x}}{\Gamma(\alpha_1(t))} dx \\
&= \int_{\beta_1 \nu_i}^{\beta_1(x_i + \nu_{i-1})} \frac{y^{\alpha_1(t)-1} e^{-y}}{\Gamma(\alpha_1(t))} dy \\
&= \int_0^{\beta_1(x_i + \nu_{i-1})} \frac{y^{\alpha_1(t)-1} e^{-y}}{\Gamma(\alpha_1(t))} dy - \int_0^{\beta_1 \nu_i} \frac{y^{\alpha_1(t)-1} e^{-y}}{\Gamma(\alpha_1(t))} dy = \\
&= \frac{\gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))}{\Gamma(\alpha_1(t))} - \frac{\gamma(\alpha_1(t), \beta_1 \nu_i)}{\Gamma(\alpha_1(t))} \\
&= \frac{\Gamma(\alpha_1(t), \beta_1 \nu_i) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))}{\Gamma(\alpha_1(t))}.
\end{aligned}$$

Therefore, the probability distribution function can be derived by the following equation:

$$\begin{aligned}
f_{V_i^*}(t) &= \frac{\partial P(X_t > \nu_i, X_t < x_i + \nu_{i-1})}{\partial t} = \frac{\partial \left[\frac{\Gamma(\alpha_1(t), \beta_1 \nu_i) - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1}))}{\Gamma(\alpha_1(t))} \right]}{\partial t} = \frac{\frac{\partial}{\partial t} \alpha_1(t)}{\Gamma(\alpha_1(t))} \\
&\times \Gamma(\alpha_1(t), \beta_1 \nu_i) [\log(\beta_1 \nu_i) - 1] - \Gamma(\alpha_1(t), \beta_1(x_i + \nu_{i-1})) [\log(\beta_1(x_i + \nu_{i-1})) - 1] \\
&+ \Gamma(\alpha_1(t)) [\log(\beta_1(x_i + \nu_{i-1})) - \log(\beta_1 \nu_i)] \\
&+ \frac{(\beta_1 \nu_i)^{\alpha_1(t)}}{(\alpha_1(t))^2} {}_2F_2(\alpha_1(t), \alpha_1(t); \alpha_1(t) + 1, \alpha_1(t) + 1; -\beta_1 \nu_i) \\
&- \frac{(\beta_1(x_i + \nu_{i-1}))^{\alpha_1(t)}}{(\alpha_1(t))^2} {}_2F_2(\alpha_1(t), \alpha_1(t); \alpha_1(t) + 1, \alpha_1(t) + 1; -\beta_1(x_i + \nu_{i-1})) \quad (\text{E.2})
\end{aligned}$$